

ECON 311

Lecture 2: Measuring the Aggregate Economy

Measuring the Macroeconomy

National Accounting

Measuring Changes Across Time

Measuring Differences Across Countries

Introduction to Long-Run Growth

Measuring the Macroeconomy

To study the macroeconomy, we need a **statistic** that summarizes the state of overall economic activity in a country.

- The most important measure of the state of the economy is **GDP**:

GDP is the market value of all final goods and services produced within a country in a given period of time

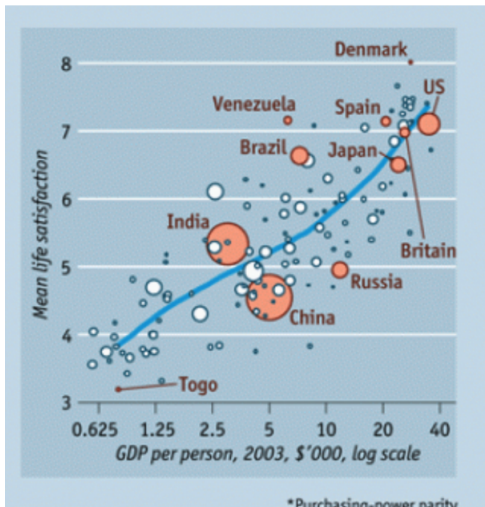
GDP per capita and well-being

Country	Real GDP per Person	Life Expectancy at Birth	HDI
Norway	\$62 448	81.6 years	0.944
United States	51 340	79.1	0.915
Canada	41 894	82.0	0.913
Japan	35 614	83.5	0.891
Russian Federation	23 564	70.1	0.798
Mexico	16 291	76.8	0.756
China	11 525	75.8	0.727
Indonesia	9 254	68.9	0.684
Philippines	6 326	68.2	0.668
India	5 238	68.0	0.609
Kenya	2 705	61.6	0.548
Ethiopia	1 336	64.1	0.442
Congo	783	58.7	0.433

Note: Data on GDP per person represent 2013 values. Real GDP per person is in 2011 dollars measured at purchasing power parity. Data on life expectancy at birth and the HDI represent 2014 values.

Source: Principles of Macroeconomics, 7ce

GDP per capita and well-being



GDP and Life Satisfaction

GDP per capita and well-being

- GDP per capita is only an imperfect measure of well-being because it does not affect several factors that affect living standards such as:
 - ▶ the value of leisure
 - ▶ the value of a clean environment
 - ▶ the value of any non-market activity

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Measuring Value of Production

There are three equivalent approaches to measuring the value of production of final goods and services:

1. Expenditure approach
2. Income approach
3. Production approach

Expenditure Approach

$$Y = C + I + G + NX$$

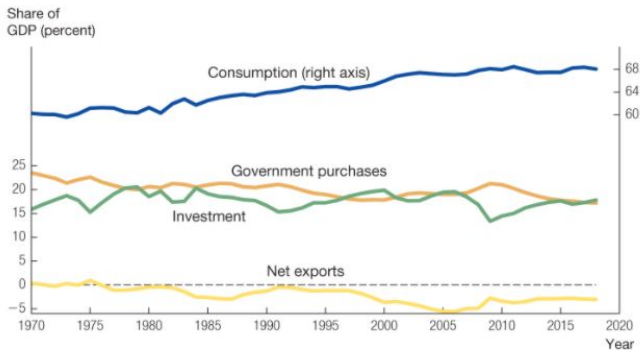
- Y =GDP (in dollars)
- C =consumption
- I =investment
- G =government purchases
- NX =exports-imports

Expenditure Approach

	Total (\$billions/year)	Per Person (\$/year)	% of Total
Gross domestic product, <i>Y</i>	\$1973	\$55 822	100%
Consumption, <i>C</i>	1107	31 311	56
Investment, <i>I</i>	478	13 530	24
Government purchases, <i>G</i>	406	11 496	21
Net exports, <i>NX</i>	-19	-527	-1
(equals exports	= (\$625	= (\$17 671	= (32%
minus imports)	- \$643)	- \$18 198)	- 33%)

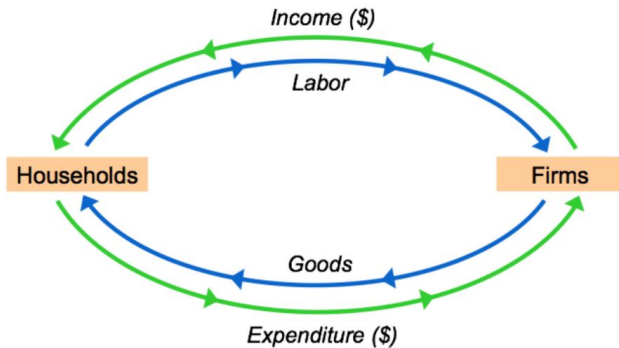
GDP in Canada 2014, expenditure approach

Expenditure Approach



Notes: Expenditure Shares of US GDP. Source: Macroeconomics, Jones

Income Approach

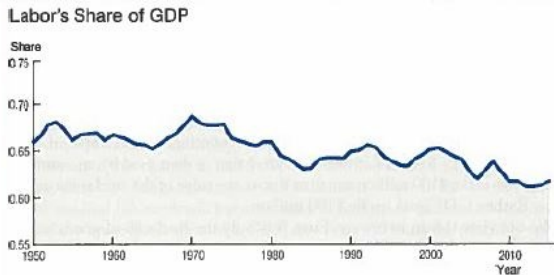


Income Approach

	Total (\$billions/year)	Per Person (\$/year)	% of Total
Gross domestic product, Y	\$1973	\$55 822	100%
Compensation of employees	999	28 253	51
Gross operating surplus	543	15 364	28
Gross mixed income	222	6 288	11
Taxes less subsidies	210	5 929	11
Statistical discrepancy	0	-11	0

GDP in Canada 2014, income approach

Income Approach



Source: Macroeconomics, Jones

Production Approach

- The production approach is based on counting the *value-added* at each stage of production
 - ▶ E.g. in a car assembly line, this approach only accounts for worker's wages but not for the value of intermediate inputs (i.e. the different parts of the car)
- In practice, it is very hard to obtain reliable *value-added* data for all sectors, ...
 - ▶ so the production approach is rarely used.

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GDP across time

We call the sum of the value of production at a particular time (t), nominal GDP:

$$NGDP_t = \sum_{i=1}^N p_{i,t} \cdot q_{i,t} \quad , \quad i = 1, \dots, N, (e.g. \ t = 2018)$$

- $\Delta NGDP$ across time happens because:
 1. Δp_i
 2. Δq_i
- To measure real changes in the economy, we need to separate 2 from 1

Quantity Indexes

Suppose we want to measure real changes between 2018 and 2019, there are two widely used quantity indexes

1. Laspeyres:

$$RGDP_{2018}^{2018\$} = \sum_{i=1}^N p_{i,2018} \cdot q_{i,2018} \quad RGDP_{2019}^{2018\$} = \sum_{i=1}^N p_{i,2018} \cdot q_{i,2019}$$

$$\% \Delta RGDP_L = \frac{RGDP_{2019}^{2018\$} - RGDP_{2018}^{2018\$}}{RGDP_{2018}^{2018\$}} * 100$$

2. Paasche:

$$RGDP_{2018}^{2019\$} = \sum_{i=1}^N p_{i,2019} \cdot q_{i,2018} \quad RGDP_{2019}^{2019\$} = \sum_{i=1}^N p_{i,2019} \cdot q_{i,2019}$$

$$\% \Delta RGDP_P = \frac{RGDP_{2019}^{2019\$} - RGDP_{2018}^{2019\$}}{RGDP_{2018}^{2019\$}} * 100$$

Quantity Indexes

- The only time $\% \Delta RGDP_L = \% \Delta RGDP_P$ is when either all prices or all quantities change proportionally, or both.
- To partially correct for this, a weighted average between the two years, called the Fisher index is implemented:

$$\% \Delta RGDP_F = \frac{\% \Delta RGDP_L + \% \Delta RGDP_P}{2}$$

Example– Calculating Chained Prices RGDP

Year	P_{apple}	Q_{apple}	$P_{computer}$	$Q_{computer}$	NGDP
2018	1	500	900	5	\$5000
2019	2	600	1000	5	\$6200
2020	3	700	1000	6	\$8100

Example– Calculating Chained Prices RGDP

Year	P_{apple}	Q_{apple}	$P_{computer}$	$Q_{computer}$	$\% \Delta RGDP_L$
2018	1	500	900	5	
2019	2	600	1000	5	2%
2020	3	700	1000	6	19.35%

Example– Calculating Chained Prices RGDP

Year	P_{apple}	Q_{apple}	$P_{computer}$	$Q_{computer}$	$RGDP_L$ Chained Prices benchmarked to 2020
2018	1	500	900	5	\$6650.9
2019	2	600	1000	5	\$6783.9
2020	3	700	1000	6	\$8100

Example– Calculating Chained Prices RGDP

Year	P_{apple}	Q_{apple}	$P_{computer}$	$Q_{computer}$	$\% \Delta RGDP_P$
2018	1	500	900	5	
2019	2	600	1000	5	3.3%
2020	3	700	1000	6	19.12%

Example– Calculating Chained Prices RGDP

Year	P_{apple}	Q_{apple}	$P_{computer}$	$Q_{computer}$	$RGDP_P$ Chained Prices benchmarked to 2020
2018	1	500	900	5	\$6583.7
2019	2	600	1000	5	\$6801
2020	3	700	1000	6	\$8100

Example– Calculating Chained Prices RGDP

Year	P_{apple}	Q_{apple}	$P_{computer}$	$Q_{computer}$	$\% \Delta RGDP_F$
2018	1	500	900	5	
2019	2	600	1000	5	2.65%
2020	3	700	1000	6	19.24%

Example– Calculating Chained Prices RGDP

Year	P_{apple}	Q_{apple}	$P_{computer}$	$Q_{computer}$	RGDP Chained Weighted Prices benchmarked to 2020
2018	1	500	900	5	\$6616.7
2019	2	600	1000	5	\$6795.3
2020	3	700	1000	6	\$8100

Price Indexes and Inflation

- To calculate price changes across time, we use price indexes:

$$GDP_t^{deflator} = \frac{NGDP_t}{RGDP_t}$$

$$Inflation_t = \frac{GDP_t^{deflator} - GDP_{t-1}^{deflator}}{GDP_{t-1}^{deflator}}$$

Example– Price Indexes and Inflation

Year	NGDP	RGDP Chained Weighted Prices benchmarked to 2020	$GDP^{deflator}$	Inflation Rate
2018	\$5000	\$6616.7	0.756	
2019	\$6200	\$6795.3	0.912	20.63%
2020	\$8100	\$8100	1	9.75%

GDP Across Time– Conclusion

- We use Quantity and Price Indexes to separate the effect of changes in real quantities from changes in prices.
- An alternative and simpler way of calculating quantity and price indexes is to use constant prices for a given year
 - ▶ Introduce problems when used over long periods in cases where
 - ▶ relative prices of goods changes because of changes in their relative quality (e.g. computers)
 - ▶ Chain Weighting solves this issue by both...
 - ▶ considering short time periods (chain)
 - ▶ combining information from both current and future prices (weight)

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Differences in Economic Activity Across Countries

- We could compare NGDP using a common currency (e.g. transform Canadian NGDP in US Dollars):

$$NGDP_{CAN}^{USD} = \frac{1}{e_{CAD/USD}} NGDP_{CAN}^{CAD}$$

- ▶ $e_{CAD/USD}$ does not perfectly reflect differences in prices across countries (e.g. differences in production costs)...
 - ▶ so comparing $NGDP_{CAN}^{USD}$ with $NGDP_{US}^{USD}$ has the same problem as comparing $NGDP$ across time
- To control for price differences, we use (PPP)

$$RGDP_{CAN}^{USD} = \frac{P_{US}}{P_{CAN}^{USD}} NGDP_{CAN}^{USD}$$

- In practice, we usually use Chain-Weighting to facilitate comparison across time

RGDP Across Countries

	Canada	United States	Mexico
GDP Growth	1.7%	1.7%	1.4%
GDP per head	\$57,190(PPP:\$67,090)	\$86,190(PPP:\$86,190)	\$14,290(PPP:\$27,060)
Inflation	2.2%	3.0%	4.0%
Budget Balance (% GDP)	-1.0	-6.9	-4.0
Population	40.1M	347.3M	131.9M

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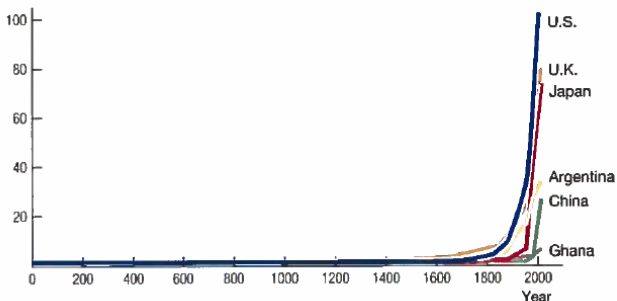
Introduction to Growth

- Standards of living in industrialized countries have greatly increased over the past few centuries.
 - ▶ E.g. Averages incomes today in North America and Western Europe are between 50 and 300 times larger than two centuries ago.
- This increase in standards of living has been a very recent phenomena in human history.
 - ▶ For most of their history (before the Industrial Revolution), humans lived barely above subsistence levels.
- Economic growth has been very uneven across time.
 - ▶ E.g. the productivity growth slowdown of 1990's
- Economic growth has been very uneven across countries.
 - ▶ E.g. growth miracles (Japan, South Korea, Singapore), growth disasters (Argentina, Sub-Saharan countries)

Economic Growth Over the Very Long Run

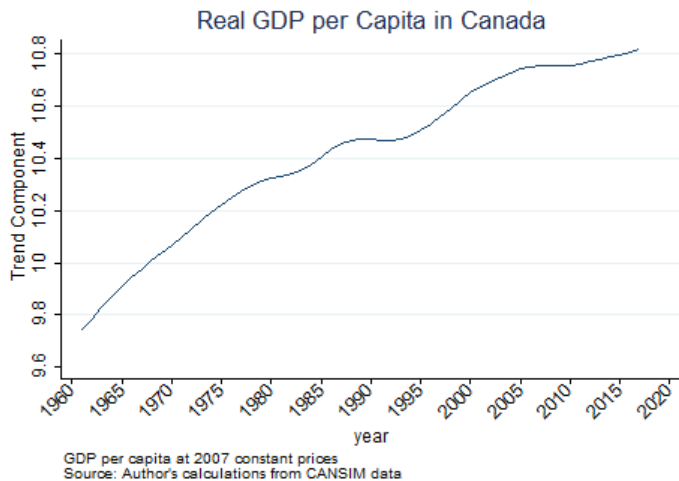
Economic Growth over the Very Long Run in Six Countries

Per capita GDP
(multiple of 300 dollars)



Source: Macroeconomics, Jones

Economic Growth Over Time



Defining Economic Growth

- Let g be a constant growth rate over a period of time, then ...

$$y_{t+T} = y_t(1 + g)^T$$

- Important properties of growth rates:

- (Rule of 70) Let $y_{t+T} = 2y_t$ and , then

$$T \approx \frac{0.7}{g}$$

- In general, the time it takes for a series to increase by a certain ratio does not depend on the initial level.
- Algebraic properties of growth rates (let the variable i grow at a constant growth rate g_i):

$$z = \frac{x}{y}$$

$$\Rightarrow g_z \approx g_x - g_y$$

$$z = x \cdot y$$

$$\Rightarrow g_z \approx g_x + g_y$$

$$z = x^a \text{ (where } a \text{ is a constant)} \Rightarrow g_z \approx a \cdot g_x$$